author date

title

Lema 1. $\forall N \in \mathbb{N} : F_N(x)$ *es* 1*-periódica. Además,*

$$\forall N \in \mathbb{N}, x \in \left[-1/2, 1/2\right) : F_N(x) = \begin{cases} \frac{1}{N}\left(\frac{\sin(\pi N x)}{\sin(\pi x)}\right)^2 & x \neq 0, \\ N & x = 0. \end{cases}$$

***Demostración*:** En primer lugar, para $x = 0$, tenemos que

$$F_N(0) = \frac{1}{N}\sum_{k=0}^{N-1} D_k(0) \overset{\substack{\texttt{lem-formula-nucleo-dirichlet/1} \\ \downarrow}}{=} \frac{1}{N}\sum_{k=0}^{N-1}(2k+1) = \frac{1}{N}[1+3+5+\cdots+(2N-3)+(2N-1)] = N.$$

Si $x \neq 0$, por el `lem-formula-nucleo-dirichlet/`lema 1, tenemos que

$$F_N(x) = \frac{1}{N}\sum_{k=0}^{N-1}\frac{\sin((2k+1)\pi x)}{\sin(\pi x)}.$$

Aplicando que $2(\sin(\pi x))(\sin((2k+1)\pi x) = \cos(2k\pi x) - \cos(2(k+1)\pi x)$ obtenemos

$$\begin{aligned} F_N(x) &= \frac{1}{2N}\frac{1}{\sin^2(\pi x)}\sum_{k=0}^{N-1}[\cos(2k\pi x) - \cos(2(k+1)\pi x)] \\ &= \frac{1}{N}\frac{1}{\sin^2(\pi x)}\frac{1-\cos(2N\pi x)}{2} = \frac{1}{N}\left(\frac{\sin(\pi N x)}{\sin(\pi x)}\right)^2. \end{aligned}$$

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